EEE104 – Digital Electronics (I)

Lecture 9

Dr. Yuxiao Fang

Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

In This Session

Boolean Algebra and Logic Simplification

SOP and POS: definitions and standard forms

Binary representation of standard terms

Boolean expressions and truth tables

Conversion between standard SOP and POS

Standard Form of Boolean Expression

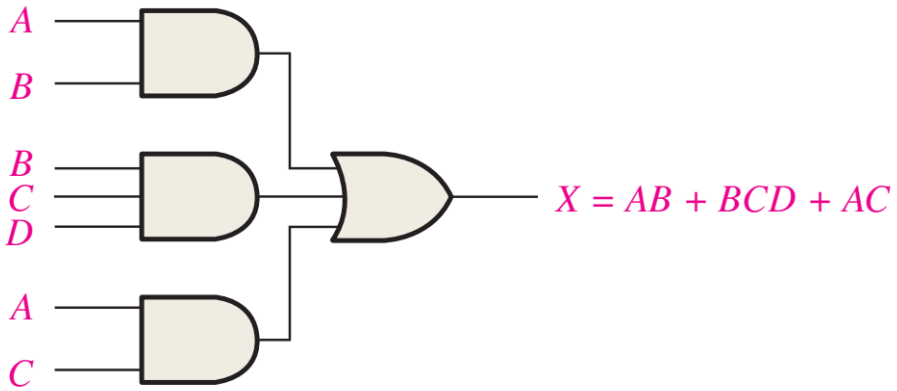
The Sum-of-Products (SOP) Form

An SOP is a sum of product terms by Boolean addition.

Example: $AB + \overline{B}C\overline{D}$

The domain is the set of variables: here A, B, C, D.

An SOP is implemented by AND-OR logic. Another example:



Standard Form of Boolean Expression

The Standard SOP Form

Every variable in the domain appears once in each product term.

$$A\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + AB\overline{C}\,\overline{D}$$

Use Rule 6 to insert a missing variable without changing the function.

Rule 6: $A + \overline{A} = 1$

$$\overline{A}\,\overline{B} + AB\overline{C}D$$

$$= \overline{A}\,\overline{B}(C + \overline{C}) + AB\overline{C}D$$

$$= \overline{A}\,\overline{B}C(D + \overline{D}) + \overline{A}\,\overline{B}\,\overline{C}(D + \overline{D}) + AB\overline{C}D$$

$$= \overline{A}\,\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D + \overline{A}\,\overline{B}\,\overline{C}\,\overline{D} + AB\overline{C}D$$

Each expansion multiplies by 1. The function stays the same.

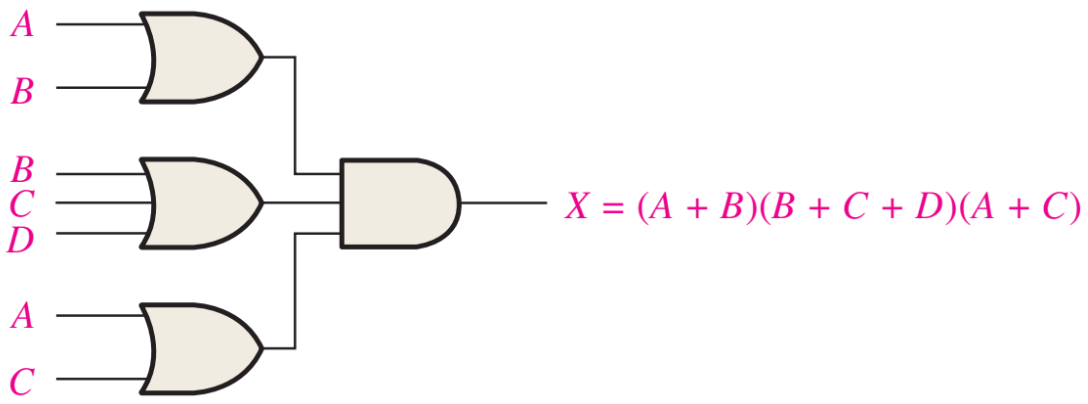
Standard Form of Boolean Expression

The Product-of-Sums (POS) Form

A POS is a product of sum terms by Boolean multiplication.

Example: $(\mathrm{A} + \mathrm{B})(\overline{\mathrm{B}} + \mathrm{C} + \overline{\mathrm{D}})$

A POS is implemented by OR-AND logic. Another example:



Standard Form of Boolean Expression

The Standard POS Form

Every variable in the domain appears once in each sum term.

$$(A + \overline{B} + C + D)(\overline{A} + \overline{B} + C + \overline{D})(A + B + \overline{C} + \overline{D})$$

Use Rule 8 (add 0) and Rule 12 (factor the expression).

Rule 8: $A \cdot \overline{A} = 0$

Rule 12: $(A + B)(A + C) = A + BC$

$$(\overline{A} + \overline{B})(A + B + \overline{C})$$

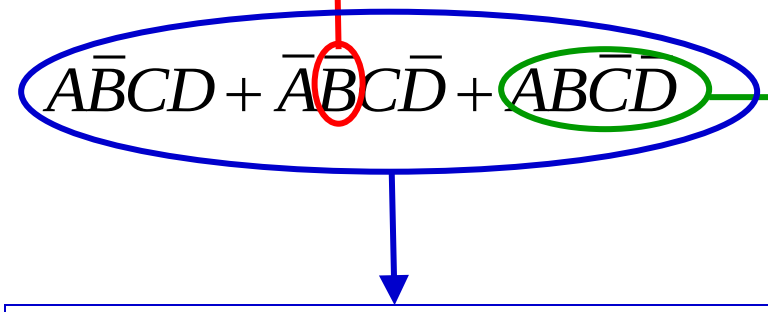
$$= (\overline{A} + \overline{B} + C\overline{C})(A + B + \overline{C})$$

$$= (\overline{A} + \overline{B} + C)(\overline{A} + \overline{B} + \overline{C})(A + B + \overline{C})$$

The added product is 0. The function stays the same.

Boolean Expressions: A Review

- A **literal** is a variable or its complement.

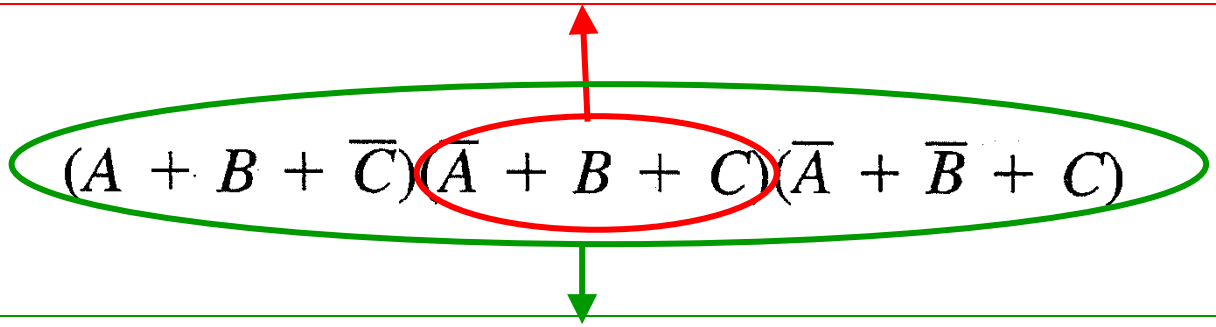


- A **product term** is the product of literals.

- A **sum-of-products (SOP)** is the sum of product terms.
- A **standard SOP** is one in which all variables appear in each product term.

Boolean Expressions: A Review

- A **sum term** is the sum of literals.



- A **product-of-sums (POS)** is the product of sum terms.
- A **standard POS** is one in which all variables appear in each sum term.

Standard Form of Boolean Expression

Binary Representation

A standard product term equals **1** for its corresponding input row.

$$\mathrm{ABCD} = 1010: \quad \mathrm{A}\overline{\mathrm{B}}\mathrm{C}\overline{\mathrm{D}} = 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

A standard sum term equals **0** for its corresponding input row.

$$\mathrm{ABCD} = 0101: \quad \mathrm{A} + \overline{\mathrm{B}} + \mathrm{C} + \overline{\mathrm{D}} = 0$$

The overbar choices differ because the required term values differ.

Practice

Domain A, B, C. Write both standard terms for input 110.

Boolean Expressions and Truth Tables

Converting SOP to a Truth Table

$$X = \overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + ABC$$

1. Convert to standard SOP.

2. Find the binary input row for each product term.

3. Set those outputs to 1. Set all other outputs to 0.

A	B	C	X
0	0	0	0
0	**0**	**1**	**1**
0	1	0	0
0	1	1	0
1	**0**	**0**	**1**
1	0	1	0
1	1	0	0
1	**1**	**1**	**1**

Boolean Expressions and Truth Tables

Converting POS to a Truth Table

$$X = (A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C}) \cdot (\overline{A} + B + \overline{C})(\overline{A} + \overline{B} + C)$$

1. Convert to standard POS.

2. Find the binary input row for each sum term.

3. Set those outputs to 0. Set all other outputs to 1.

A	B	C	X
0	**0**	**0**	**0**
0	0	1	1
0	**1**	**0**	**0**
0	**1**	**1**	**0**
1	0	0	1
1	**0**	**1**	**0**
1	**1**	**0**	**0**
1	1	1	1

Boolean Expressions and Truth Tables

Determine SOP Expressions from Truth Tables

1. Select the input rows for which the output is 1.
2. Write a product term for each row, then OR the terms.

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	**1**	**1**	**1**
1	**0**	**0**	**1**
1	0	1	0
1	**1**	**0**	**1**
1	**1**	**1**	**1**

$011 \rightarrow \overline{A}BC$

$100 \rightarrow A\overline{B}\,\overline{C}$

$110 \rightarrow AB\overline{C}$

$111 \rightarrow ABC$

$X = \overline{A}BC + A\overline{B}\,\overline{C} + AB\overline{C} + ABC$

Boolean Expressions and Truth Tables

Determine POS Expressions from Truth Tables

1. Select the input rows for which the output is 0.
2. Write a sum term for each row, then AND the terms.

A	B	C	X
0	**0**	**0**	**0**
0	**0**	**1**	**0**
0	**1**	**0**	**0**
0	1	1	1
1	0	0	1
1	**0**	**1**	**0**
1	1	0	1
1	1	1	1

$$000 \rightarrow (A + B + C)$$

$$001 \rightarrow (A + B + \overline{C})$$

$$010 \rightarrow (A + \overline{B} + C)$$

$$101 \rightarrow (\overline{A} + B + \overline{C})$$

$$X = (A + B + C)(A + B + \overline{C}) \cdot (A + \overline{B} + C)(\overline{A} + B + \overline{C})$$

Standard Form of Boolean Expression

Converting Standard SOP to Standard POS

$$X = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}C + ABC$$

1. Determine the input rows represented by the product terms.

000, 010, 011, 101, 111

2. Find the remaining rows in the fixed domain A, B, C.

001, 100, 110

3. Write the corresponding sum terms and multiply them.

$$X = (A + B + \overline{C})(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$$

Same function. Reverse the row selection to convert POS back to SOP.

SOP and POS: Key Differences

	SOP	POS
Inside each term	AND	OR
Between terms	OR	AND
Condition to identify	**A product term = 1**	**A sum term = 0**
Output then equals	**1**	**0**
Rows used from a table	**The 1 rows**	**The 0 rows**

Standard form: every term contains every variable in the domain once.

SOP/POS describes the structure; “standard” adds a separate requirement.

Practice: Standard SOP and POS

A	B	C	H
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Use the domain A, B, C.

1. Write the standard SOP.

2. Write the standard POS.

Next: **Karnaugh maps** arrange these input combinations for simplification.